

Surface-initiated ARGET ATRP of zwitterionic carboxybetaine methacryloyl ethylenediamine in formation of ultralow-biofouling films

Inae Han  | HyeongBin Rheem  | Sang Yeong Han | Jayeong Kim |
Insung S. Choi 

Department of Chemistry, KAIST, Daejeon, Korea

Correspondence

Insung S. Choi, Department of Chemistry, KAIST,
Daejeon 34141, Korea.
Email: ischoi@kaist.ac.kr

Funding information

National Research Foundation of Korea,
Grant/Award Numbers: RS-2024-00335713,
2021R1A3A3002527

Abstract

Undesired biofouling processes severely compromise the performance and durability of materials, including those used in biomedical and marine environments, strongly demanding effective and robust antifouling coating strategies. This paper presents the design, synthesis, and application of a highly water-soluble zwitterionic monomer, carboxybetaine methacryloyl ethylenediamine (CB-MAEDA), in the fabrication of ultralow-biofouling polymer brushes via surface-initiated, activators regenerated by electron transfer atom transfer radical polymerization (SI-ARGET ATRP). CB-MAEDA is synthesized through a reproducible, high-yield (>80%) route, and its surface-initiated polymerization under ambient aqueous conditions produces uniform poly(CB-MAEDA) brushes, which achieve a 96.5% reduction in *Escherichia coli* adhesion. Collectively, these results demonstrate that SI-ARGET ATRP of CB-MAEDA affords thick, highly hydrophilic, and exceptionally effective antifouling coatings, underscoring its potential as a versatile platform for interface engineering in biomedical and related applications.

KEYWORDS

antifouling, carboxybetaine, SI-ARGET ATRP, zwitterions

INTRODUCTION

Biofouling and biofilm formation pose great challenges across various sectors, from medical implants and prosthetic devices¹ to the marine industry.² In marine environments, biofouling typically progresses through four stages: initial adhesion of biomolecules (film formation), colonization by microorganisms such as bacteria (primary adhesion), attachment of diatoms and microalgae (secondary adhesion), and settlement of larger organisms including barnacles and mussels (tertiary adhesion).³ Among various antifouling strategies, zwitterionic polymers have emerged as a particularly promising class of materials in addressing these issues.⁴ Zwitterionic moieties, such as phosphorylcholine (PC), sulfobetaine (SB),

and carboxybetaine (CB), have well-hydrated structures, thus displaying high resistance to the adhesion of marine biomolecules and microorganisms. Surface functionalization with zwitterionic polymers has been achieved mainly by two methods: “grafting-to” and “grafting-from.”^{5,6} In the grafting-to approach, pre-synthesized, end-functionalized polymers are directly attached to the substrate surface. By contrast, the grafting-from approach (aka, surface-initiated polymerization, SIP) enables the formation of higher surface-packing densities and affords precise control over polymer-chain length.

Since Jiang's pioneering work,⁷ surface-initiated atom transfer radical polymerization (SI-ATRP) of zwitterionic monomers has widely been employed for the fabrication of anti-biofouling films. However, the conventional SI-ATRP suffers from several limitations, including the need for rigorous oxygen removal and the substantial use

Inae Han and HyeongBin Rheem contributed equally.

In memory of Professor Hyun-Joon Ha.

of potentially toxic catalysts, which hinder its practical applicability.^{7–9} To overcome these challenges, an advanced strategy known as activators regenerated by electron transfer ATRP (ARGET ATRP) was developed.¹⁰ In ARGET ATRP, a reducing agent (e.g., L-ascorbic acid) continuously regenerates active Cu(I) species from inactive Cu(II) species in an aqueous solution under mild, ambient conditions, minimizing the overall concentration of Cu species.^{11,12} The mild nature of ARGET ATRP has enabled surface engineering of living cells.^{13–15}

To date, various zwitterionic monomers have been employed in SI-ARGET ATRP to fabricate anti-biofouling films. CB acrylamide films, for example, effectively inhibit the nonspecific adsorption of proteins, though its antifouling efficacy against microfouling organisms has yet to be demonstrated.^{16–18} In another study, after immobilizing the 3-(trimethoxysilyl)propyl 2-bromo-2-methylpropionate initiator onto polydopamine layers, zwitterionic PC, SB, and CB polymer brushes were fabricated to achieve anti-biofouling against proteins.¹⁹ Furthermore, the lysine-based zwitterionic polymer brushes, containing a quaternary ammonium group positioned near the carboxyl group showed strong anti-biofouling performance against proteins and bacteria.²⁰ However, even after adjusting the monomer ratio, the maximum thickness of grafted polymers remained below 10 nm, which is generally considered the minimum threshold for effective anti-biofouling performance.^{21,22} In addition, although certain lysine-based monomers are soluble in water,²³ the practical necessity of using organic solvents, such as dimethylformamide or ethanol, for specific derivatives remains a challenge for broad applications.^{20,24} In this work, we designed and synthesized CB methacryloyl ethylenediamine (denoted CB-MAEDA) as a monomer for the fabrication of ultralow-biofouling films via SI-ARGET ATRP (Figure 1a). The fabricated poly(CB-

MAEDA) brushes demonstrated several advantages, including the high water solubility of monomers, increased polymer brush thickness, and a 96.5% reduction in *Escherichia coli* adhesion.

The synthetic route for CB-MAEDA is illustrated in Figure 1b. Briefly, 2-aminoethyl methacrylamide hydrochloride (AEMA, **1**) was synthesized following a previously reported protocol through the nucleophilic substitution reaction between methacryloyl chloride and ethylenediamine (Figure S1 for the ¹H and ¹³C NMR spectra).²⁵ The amine functionality of the compound **1** provided a reactive site for subsequent amidation with the carboxylic acid group of *N*-Boc-L-glutamic acid 1-*tert*-butyl ester (Boc-Glu-O^tBu), yielding the compound **2** (Figure S2).²⁶ After the removal of the Boc and *tert*-butyl protecting groups, the intermediate compound **3** was obtained (Figure S3) and subjected to exhaustive methylation using iodomethane. The reaction afforded the desired zwitterionic CB derivative, CB-MAEDA, which contained a quaternary ammonium group adjacent to a carboxylate, effectively mimicking the CB structure (Figure S4).^{20,27} This synthetic route was highly reproducible and provided CB-MAEDA in good overall yield (>80%).

Poly(CB-MAEDA) brushes were fabricated on gold substrates via SI-ARGET ATRP. Prior to polymerization, the gold substrates were functionalized with the self-assembled monolayers (SAMs) of ω -mercaptoundecyl bromoisobutyrate, which served as an initiator layer.²⁸ After the formation of the initiator SAMs, the static water contact angles of the gold substrates decreased from 66.8° to 56.7° (Figure S5a). In addition, the X-ray photoelectron spectroscopy (XPS) spectrum of the SAMs exhibited a Br 3d peak at approximately 68 eV, further supporting the successful formation of the initiator SAMs (Figure S5b). Subsequent SI-ARGET ATRP was carried out

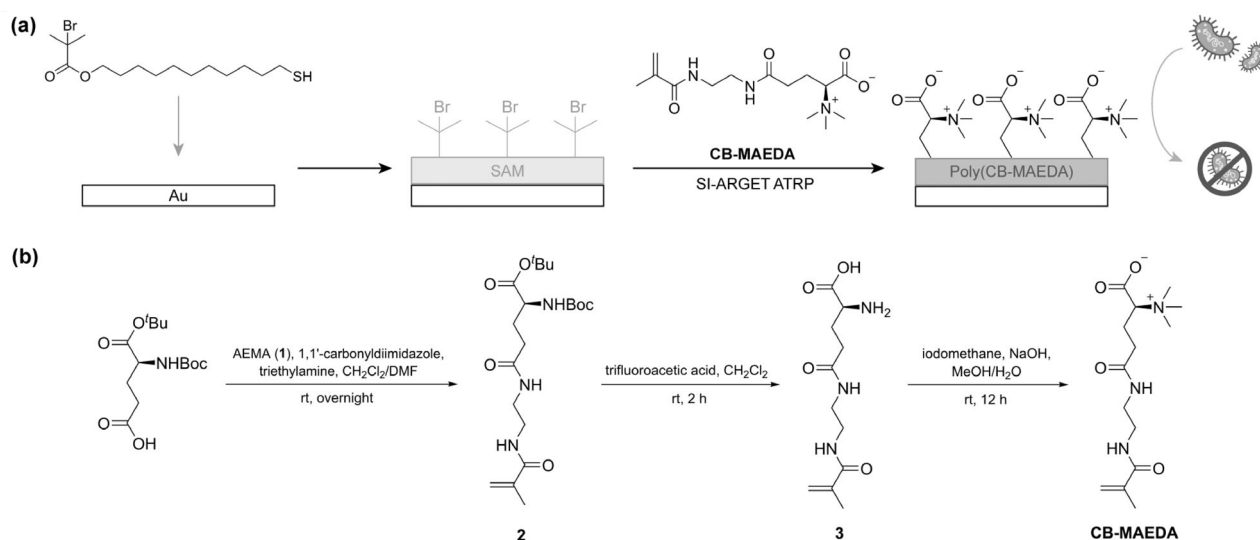


FIGURE 1 (a) Schematic for the formation of ultralow-biofouling films via SI-ARGET ATRP of carboxybetaine methacryloyl ethylenediamine (CB-MAEDA). SAM, self-assembled monolayer. (b) Synthetic scheme for CB-MAEDA.

in an aqueous solution under ambient air, employing an ultralow concentration of Cu(II) catalyst and L-ascorbic acid as the reducing agent.^{20,29,30} In brief, the reaction mixture contained 600 mM CB-MAEDA, 0.01 mM copper(II) bromide (CuBr_2), 1 mM 2,2'-bipyridyl, and 3 mM L-ascorbic acid. Ellipsometric measurements indicated that the film thickness was 3.7 nm after 3 h of reaction and gradually increased to 23.0 and 31.1 nm after 15 and 24 h, respectively (Figure S6). A previous study reported that zwitterionic poly(CB methacrylate) brushes required a minimum thickness of 10–15 nm to achieve ultralow nonspecific protein adsorption.²¹ Another study indicated that the optimal thickness of poly(CB methacrylate) films for antifouling performance ranged between 20 and 30 nm.²² Based on these reports, a polymerization time of 15 h was selected to obtain an appropriate film thickness.

After the SI-ARGET ATRP process, the static water contact angles of the substrate decreased markedly from 56.7° to 11.5° , indicating the formation of a highly hydrophilic surface (Figure S5a). This hydrophilicity was attributed to the strong hydration of the zwitterionic CB groups, which are known to form tightly bound water layers that effectively inhibit nonspecific protein and bacterial adsorption.³¹ The XPS spectrum exhibited characteristic peaks corresponding to C 1s (284.4 eV), N 1s (398.6 eV), and O 1s (531.1 eV), which were attributed to the alkyl chains, quaternary ammonium, and carboxylate groups of the poly(CB-MAEDA), respectively (Figure 2a). Given that nanofilms with thicknesses of tens of nanometers typically attenuate signals from underlying substrates,³² the disappearance of the Au 4p, 4d, and 4f peaks, as well as the S 2p peak, originating from the SAMs, further confirmed that the gold substrate was fully covered by the poly(CB-MAEDA) film. The high-resolution N 1s spectrum displayed two deconvoluted components at 402.5 and 399.6 eV, corresponding to quaternary ammonium and secondary amine, respectively, consistent with the chemical structure of poly(CB-MAEDA). Furthermore, the Fourier-transform infrared (FT-IR) spectra showed several characteristic absorption bands indicative of successful SAM formation and SI-ARGET ATRP (Figure 2b). Distinct peaks at 2923 and 2852 cm^{-1} were assigned to the asymmetric and symmetric stretching modes of methylene groups, while the peak at 1455 cm^{-1} corresponded to the bending mode of aliphatic carbons in the SAM. Additionally, the strong bands appearing at 1737, 1280, and 1174 cm^{-1} were attributed to the C=O stretching and the asymmetric/symmetric C—O stretching vibrations, respectively. These data consistently supported the successful formation of the SAM on the gold substrates. Following SI-ARGET ATRP, a broad band centered at 3463 cm^{-1} corresponded to N—H stretching vibration of secondary amides, while strong peaks at 1643 and 1542 cm^{-1} were attributed to the C=O stretching (Amide I) and N—H bending (Amide II) of amide groups (Figure 2b). Distinct bands at 1596 and 1367 cm^{-1}

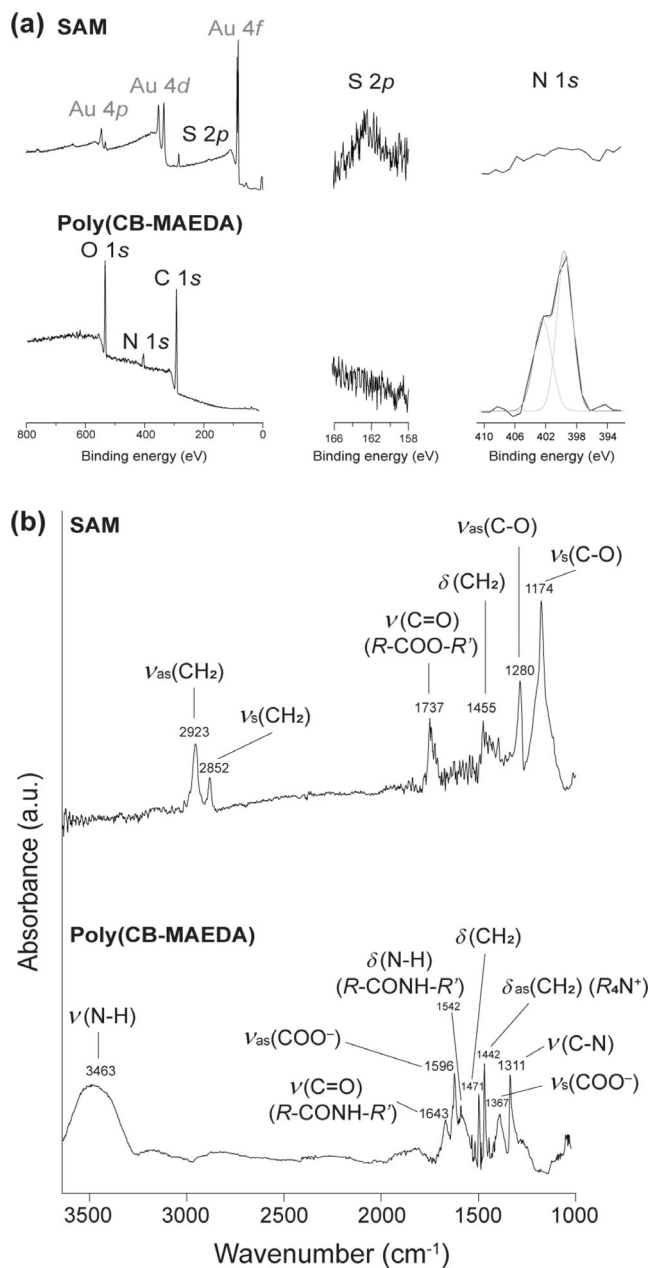


FIGURE 2 (a) XPS and (b) FT-IR spectra of the initiator SAMs and poly(CB-MAEDA) films.

were assigned to the asymmetric and symmetric stretching of the carboxylate groups, respectively. The peaks at 1471 and 1442 cm^{-1} were ascribed to CH_2 bending and quaternary ammonium vibrations, confirming the presence of the zwitterionic CB moieties. In addition, the peak at 1311 cm^{-1} was assigned to C—N stretching, further supporting the incorporation of secondary and quaternary amine functionalities. These spectral features collectively verified the successful formation of poly(CB-MAEDA) brushes.

To evaluate the antifouling properties of the poly(CB-MAEDA) films, *E. coli* was chosen as a model microorganism, since it is one of the most common biofilm-forming

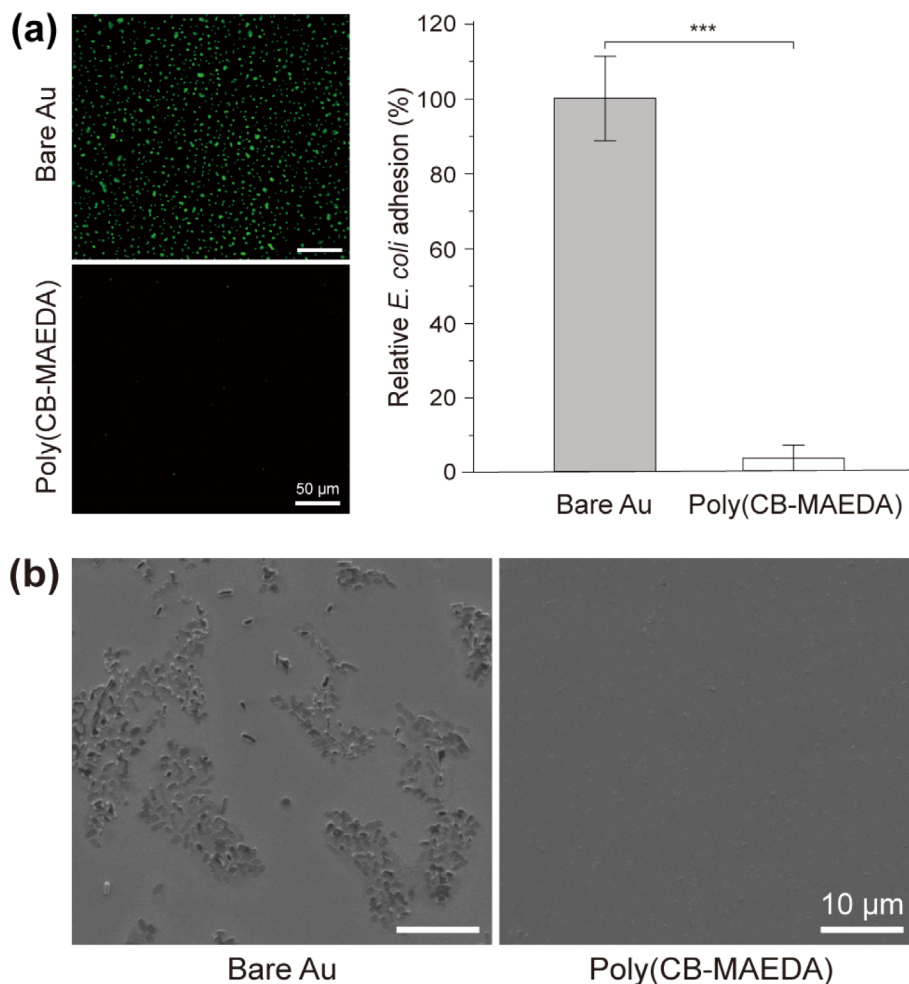


FIGURE 3 (a) CLSM and (b) FE-SEM images of bare gold and poly(CB-MAEDA) substrates after incubation with *Escherichia coli*. Statistical significance was analyzed by unpaired Student's *t*-test (** $p < 0.001$).

species associated with infections on medical devices such as catheters and implants.^{33,34} After incubation of bare and poly(CB-MAEDA)-coated gold substrates in *E. coli* suspension for 6 h, confocal laser-scanning microscopy (CLSM) was employed to visualize bacterial adhesion. As shown in Figure 3a, the bare gold substrate exhibited extensive bacterial attachment, forming dense microcolonies, whereas the poly(CB-MAEDA)-coated surface showed markedly reduced bacterial adhesion. Quantitative image analysis revealed that the relative bacterial coverage on the poly(CB-MAEDA)-coated substrate was only 3.5% compared with that of the bare gold. Field-emission scanning electron microscopy (FE-SEM) further supported these observations, showing extensive bacterial adhesion and clustering on the bare surface, while bacteria were nearly absent on the poly(CB-MAEDA)-coated surface (Figure 3b). The antifouling efficacy observed in this study was consistent with the previous reports on CB-based coatings, which have demonstrated over 80% reductions in bacterial adhesion compared with unmodified substrates.^{32,35}

CONCLUSION

In summary, we designed and synthesized carboxybetaine methacryloyl ethylenediamine (CB-MAEDA) and successfully utilized it to fabricate ultralow-biofouling films via surface-initiated ARGET ATRP. The resulting poly(CB-MAEDA) films exhibited remarkable resistance to bacterial adhesion, with a 96.5% reduction in *E. coli* attachment compared with bare substrates. These results indicate the strong potential of CB-MAEDA as a versatile precursor for the development of effective anti-biofouling coatings, particularly for biomedical and related applications. Moreover, we anticipate that the characteristics of CB-MAEDA, including its high water solubility, will enable its seamless application in bio-related fields, such as single-cell nanoencapsulation (SCNE).^{15,36,37}

ACKNOWLEDGMENTS

This work was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) (RS-2024-00335713 and 2021R1A3A3002527).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Inae Han  <https://orcid.org/0009-0002-6336-4242>

HyeongBin Rheem  <https://orcid.org/0009-0008-9166-6882>

Insung S. Choi  <https://orcid.org/0000-0002-9546-673X>

REFERENCES

- [1] C. M. C. Faustino, S. M. C. Lemos, N. Monge, I. A. C. Ribeiro, *Adv. Colloid Interface Sci.* **2020**, *284*, 102230.
- [2] P. Vuong, A. McKinley, P. Kaur, *NPJ Mater. Degrad.* **2023**, *7*, 50.
- [3] S. Tian, Y. Li, H. Zhang, G. Lu, R. Li, J. Yu, C. Zhao, J. Yang, L. Zhang, *Langmuir* **2025**, *41*, 1037.
- [4] Z. Huang, J. Chen, R. Li, S. Shi, Y.-K. Gong, *Prog. Org. Coat.* **2023**, *182*, 107636.
- [5] Y. Liu, V. Klep, B. Zdyrko, I. Luzinov, *Langmuir* **2005**, *21*, 11806.
- [6] B. Li, B. Yu, Q. Ye, F. Zhou, *Acc. Chem. Res.* **2015**, *48*, 229.
- [7] Z. Zhang, S. Chen, Y. Chang, S. Jiang, *J. Phys. Chem. B* **2006**, *110*, 10799.
- [8] B. S. Lee, Y. S. Chi, K.-B. Lee, Y.-G. Kim, I. S. Choi, *Biomacromolecules* **2007**, *8*, 3922.
- [9] G. Szczepaniak, L. Fu, H. Jafari, K. Kapil, K. Matyjaszewski, *Acc. Chem. Res.* **2021**, *54*, 1779.
- [10] W. Jakubowski, K. Matyjaszewski, *Angew. Chem., Int. Ed.* **2006**, *45*, 4482.
- [11] B. S. Lee, J. Y. Kim, J. H. Park, W. K. Cho, I. S. Choi, *J. Nanosci. Nanotechnol.* **2016**, *16*, 3106.
- [12] A. Simakova, S. E. Averick, D. Konkolewicz, K. Matyjaszewski, *Macromolecules* **2012**, *45*, 6371.
- [13] J. Y. Kim, B. S. Lee, J. Choi, B. J. Kim, J. Y. Choi, S. M. Kang, S. H. Yang, I. S. Choi, *Angew. Chem., Int. Ed.* **2016**, *55*, 15306.
- [14] Y. Xue, Y.-X. Chen, Y.-Y. Yu, Y.-C. Yong, *Chem. Eng. J.* **2020**, *387*, 124068.
- [15] H. B. Rheem, N. Kim, D. T. Nguyen, G. A. Baskoro, J. H. Roh, J. K. Lee, B. J. Kim, I. S. Choi, *Chem. Rev.* **2025**, *125*, 6366.
- [16] D. Hong, H.-C. Hung, K. Wu, X. Lin, F. Sun, P. Zhang, S. Liu, K. E. Cook, S. Jiang, *ACS Appl. Mater. Interfaces* **2017**, *9*, 9255.
- [17] Z. Zhang, H. Vaisocherová, G. Cheng, W. Yang, H. Xue, S. Jiang, *Biomacromolecules* **2008**, *9*, 2686.
- [18] E. van Andel, S. C. Lange, S. P. Pujari, E. J. Tijhaar, M. M. J. Smulders, H. F. J. Savelkoul, H. Zuilhof, *Langmuir* **2019**, *35*, 1181.
- [19] N. Li, T. Li, X.-Y. Qiao, R. Li, Y. Yao, Y.-K. Gong, *ACS Appl. Mater. Interfaces* **2020**, *12*, 12337.
- [20] J. Lv, J. Jin, J. Chen, B. Cai, W. Jiang, *ACS Appl. Mater. Interfaces* **2019**, *11*, 25556.
- [21] T. Sentoukas, W. Walach, K. Filipek, B. Trzebicka, *Materials* **2025**, *18*, 4514.
- [22] G. Cheng, G. Li, H. Xue, S. Chen, J. D. Bryers, S. Jiang, *Biomaterials* **2009**, *30*, 5234.
- [23] J. Liu, Y. Tang, X. Lin, C. Zhang, N. Ma, L. Wang, W. Qu, M. Shao, Y. Zhao, W. Zhang, *ACS Appl. Mater. Interfaces* **2025**, *17*, 66505.
- [24] P. Dinda, M. Anas, P. Banerjee, T. K. Mandal, *Macromolecules* **2022**, *55*, 4011.
- [25] G. Y. N. Chan, A. G. Jhingran, P. A. Kambouris, M. G. Looney, D. H. Solomon, *Polymer* **1998**, *39*, 5781.
- [26] Q. Liu, W. Li, H. Wang, B.-m. Z. Newby, F. Cheng, L. Liu, *Langmuir* **2016**, *32*, 7866.
- [27] V. N. Chary, C. D. Kumar, M. Vairamani, S. Prabhakar, *J. Mass Spectrom.* **2012**, *47*, 79.
- [28] D. M. Jones, A. A. Brown, W. T. S. Huck, *Langmuir* **2002**, *18*, 1265.
- [29] S. Y. Kim, H. J. Seo, S. Kim, W. K. Cho, *Bull. Korean Chem. Soc.* **2021**, *42*, 761.
- [30] L. B. Veldscholte, S. de Beer, *ACS Appl. Polym. Mater.* **2023**, *5*, 7652.
- [31] G. Wang, L. Wang, W. Lin, Z. Wang, J. Zhang, F. Ji, G. Ma, Z. Yuan, S. Chen, *ACS Appl. Mater. Interfaces* **2015**, *7*, 16938.
- [32] I. Han, S. Y. Kim, S.-P. Hong, I. S. Choi, W. K. Cho, *Prog. Org. Coat.* **2023**, *184*, 107860.
- [33] H. Yu, L. Liu, X. Li, R. Zhou, S. Yan, C. Li, S. Luan, J. Yin, H. Shi, *Chem. Eng. J.* **2019**, *360*, 1030.
- [34] P. Li, Z. Ding, Y. Yin, X. Yu, Y. Yuan, M. B. Pérez, S. de Beer, G. J. Vancso, Y. Yu, S. Zhang, *Eur. Polym. J.* **2020**, *134*, 109845.
- [35] J. Zhang, L. Chen, L. Chen, S. Qian, X. Mou, J. Feng, *Carbohydr. Polym.* **2021**, *257*, 117627.
- [36] S. Y. Han, Y. Jeong, H. Lee, I. S. Choi, *Bull. Korean Chem. Soc.* **2025**, *46*, 1088.
- [37] S. Park, H. Bisht, S. Park, J. Jeong, Y. Hong, D. Chu, M. Koh, D. Hong, *Macromol. Biosci.* **2025**, *25*, 2400616.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: I. Han, H. Rheem, S. Y. Han, J. Kim, I. S. Choi, *Bull. Korean Chem. Soc.* **2026**, *1*.
<https://doi.org/10.1002/bkcs.70111>